Current Clarity Stage 17: Multi-Scene Consensus

Development only.

- The single-scene engine is unchanged: C-gbm and its scene predictions are not refitted, and fold models are used exactly as in Stages 14–16.
- No field report, Stage 16 fresh row or Stage 14/15 certification row was used.
- Production is unchanged, and nothing was deployed.
- Water Temperature is not started.

Step	UTC	What was fixed
Pre-registration	2026-10-01T18:19:29Z	Windows, the 3-nearest cap, rainfall compatibility, 4 methods, S2C variants, the Strong agreement rule, selection, bars (`PREREGISTRATION_STAGE17.md`, `a058f476…`)
Inputs	2026-10-01T23:54:18Z	2,072 neighbour-scene ACOLITE results, ERA5 rainfall, the neighbour list (`scenes/`)

The answer

Combining neighbouring satellite views does help, but under the rule I froze it is not adopted.

- **What combining gains.** Weighting each frozen scene prediction by its uncertainty, over ±10 days (M-iuw), raises stained water from **0.571 to 0.667** (+0.096, CI +0.054 to +0.143) and the operational score from 0.693 to 0.743.
- **The Stage 16 freeze bars.** With it, every core class clears 0.60, so it **meets the Stage 16 freeze bars on development** for the first time.
- **What else it improves.**
  - Stained answers are right more often: 79%, from 72%.
  - Stained harm falls from 22% to 18.5%.
  - Exact feet tighten from ×1.36 to ×1.33.
- **The cost.** It answers about 100 rows that Stage 16 left Unknown, and some of those answers are wrong. The overall harmful rate rises **from 10.6% to 11.6%**. The CI of that change (–0.2 to +2.1 points) includes zero.
- **The frozen rule.** It admitted only variants that do not raise the harmful rate at all. That leaves only agreement-only variants, which keep every Stage 16 answer, so **MULTI-SCENE CONSENSUS USEFUL is NO**.

Two pre-registered ideas do not hold up:

- **The rainfall rule makes every method worse.** ERA5 point rainfall discards scenes that were in fact still informative.
- **Strong evidence reaches at most 24%,** at 95% right. That is far from the 50% target.

Sentinel-2C cannot be tested on development: only 49 of the 2,072 neighbour scenes are S2C.

Bars. The true 9/10 bar is far away. Readiness to move to Water Temperature is **NO**.

The decision this leaves you. Accepting about one extra harmful answer per hundred, a change that is not statistically significant, would buy a material stained-water gain. That needs a new, unseen certification pool either way.

1. How the consensus was built

Which scenes. Each development event gets up to its **3 nearest valid neighbouring passes** within ±10 days. These are the Stage 11 production-gated passes with ≥ 5 valid water pixels, excluding the matched pass.

- **The run.** 2,487 neighbour scenes for 1,215 events, run through the unchanged K4 ACOLITE path: 2,072 L1C items, 15 failures (6 had no L1C).
- **The predictor.** Each scene is predicted by the fold C-gbm trained without its lake.

Window	Events with ≥ 1 valid neighbour (Stage 11 production gates)
±3 days	623 of 1,582 (39%)
±5 days	919 (58%)
±7 days	1,064 (67%)
±10 days	1,215 (77%)

How each scene is weighted. Each scene carries its prediction, tier, platform and Δt. Its variance is $\sigma^2_tier + \kappa \cdot |\Delta t|$, fitted on training lakes:

- $\sigma \approx 0.19$ (tier 1) and 0.17–0.20 (tier 2);
- $\kappa \approx 0.0015$–0.0020 per day.

That growth is small: a scene 10 days away is only about 20% noisier than the matched one.

Calibration. Posteriors are re-calibrated (nested) by the number of contributing scenes, and the frozen Stage 16 gate (0.70 / 0.70) issues the state.

2. The four methods (±10 days, no rain rule, S2C off)

Development (whole-lake held out)	Operational	Harmful	Severe	Clear	Normal	Stained	Limited	Strong share / right
Stage 16 (anchor scene only)	0.693	10.6%	1.5%	0.880	0.643	0.571	0.676	11% / 97%
M-agree (±10 d, no rain rule, S2C off)	0.693	10.6%	1.5%	0.880	0.643	0.571	0.676	21% / 95%
M-iuw (±10 d, no rain rule, S2C off)	0.743	11.6%	1.3%	0.893	0.701	0.667	0.710	23% / 95%
M-robust (±10 d, no rain rule, S2C off)	0.743	11.6%	1.3%	0.892	0.701	0.667	0.710	23% / 95%
M-bayes (±10 d, no rain rule, S2C off)	0.745	11.4%	1.3%	0.893	0.703	0.671	0.711	23% / 95%
M-bayes (±10 d, no rain rule, S2C agree)	0.745	11.3%	1.4%	0.891	0.704	0.675	0.709	24% / 95%

- **Combining matters, the method does not.** The three combining methods are within 0.003 of one another. Down-weighting outliers (M-robust) and a population prior (M-bayes) add nothing over plain inverse-uncertainty weighting.
- **The lesson of Stage 11 holds.** Combining helps because it averages several independent errors. It does not help by discarding the matched scene.

3. Windows and hydrology

M-iuw, S2C off	±3 d	±5 d	±7 d	±10 d
Operational (no rain rule)	0.696	0.714	0.722	0.743
Stained (no rain rule)	0.545	0.590	0.611	0.667
Harmful (no rain rule)	11.4%	12.1%	12.3%	11.6%
Operational (rain rule)	0.696	0.707	0.715	0.726
Stained (rain rule)	0.548	0.580	0.603	0.634
Harmful (rain rule)	11.4%	12.5%	12.5%	12.3%

- **The widest window is best.** Most events have a valid neighbour only beyond ±3 days.
- **The rainfall rule (≥ 20 mm, or a day ≥ 12.7 mm) lowers the score at every window.** At the scale of an ERA5 cell, rain is a poor proxy for whether a lake's clarity changed. Hydrology may still decide compatibility, but this rule does

not earn its place.

4. Stained water, the primary question

Stained water (development)	Stage 16	Selected (agreement-only)	M-iuw ±10 d (not adopted)
Operational score	0.571	0.571	**0.667**
Answer right (full-authority answers)	72%	72%	**79%**
Harmful (stained rows)	22.0%	22.0%	**18.5%**
Mean posterior of issued states (all)	0.805	0.805	**0.830**
Exact-feet error, rows with a consensus	×1.36	×1.36	**×1.33**

M-iuw ±10 d against Stage 16 (lake bootstrap, 2,000, seed 17)	Difference	95% CI
Stained score	+0.096	+0.054 to +0.143
Class-balanced operational	+0.050	+0.035 to +0.066
Harmful rate	+0.95 pts	-0.19 to +2.08 pts

- **For stained water, every measure improves.** That includes the stained harmful rate, which falls.
- **Where the overall harm increase comes from.** About 100 former Unknowns now get answers (Unknown falls from 13% to 6%), and a few of those are wrong.

5. Strong evidence

Agreement-only (M-agree, S2C off), Strong share / right	±3 d	±5 d	±7 d	±10 d
rain rule	11% / 94%	15% / 95%	17% / 95%	19% / 95%
no rain rule	11% / 94%	17% / 95%	19% / 95%	21% / 95%

Evidence level (share of all answers)	Stage 16	Selected (agreement-only)	M-iuw ±10 d
Strong	10%	9%	22%
Moderate	70%	71%	68%
Weak	7%	7%	4%
Unavailable	13%	13%	6%
Strong right	97%	94%	95%

What agreement buys.

- **The rule:** at least 2 compatible full-authority scenes whose states contain the answer, no disagreeing scene, and a posterior ≥ 0.90.
- **What it gives:** at ±10 days, Strong rises from Stage 16's 11% to 21–24% of usable answers, at 95% right. With only ±3 days (the selected variant) it stays at 11%.
- **The target is out of reach.** It is not close to 50%: half of events have at most one valid neighbour (23% have none), and agreement often still leaves the posterior below 0.90.
- **Why not loosen it.** Lowering the bar to hit 50% would trade accuracy for coverage, which the pre-registration forbids.

6. Sentinel-2C

S2C AGREEMENT CAN UPGRADE AUTHORITY: NO.

- **The test.** S-agree against S-off, at the selected variant.
- **The result.** S2C raised **0** answers' evidence level, against the 30 required. Development contains almost no S2C: 49 of 2,072 neighbour scenes, and 38 anchors.
- **What it would take.** Adding agreeing S2C scenes changes the consensus very little (M-bayes S2C-agree against S2C-off: 0.745 against 0.745). The question can only be answered on 2025–2026 data, where S2C is a third of scenes.
- **No S2C scene regains authority alone,** and no S2C bias correction was fitted.

7. What the user would see (logic first)

Underlying evidence	Display
Strong: ≥ 2 compatible full-authority views agree, posterior ≥ 0.90	**2.7 ft · Normal-to-Stained** · High confidence — recent satellite views agree
Moderate: one view, or views that do not all agree but none disagree	**2.7 ft · Normal-to-Stained**
A compatible full-authority view disagrees	**~2.7 ft · Normal-to-Stained** · Lower confidence — recent views disagree
Weak (Sentinel-2C only, or tier 3)	**~2.7 ft · Normal-to-Stained** · Lower confidence
Unknown	Visibility unknown — no reliable recent view

- **No scene counts.** The confidence line already says what the count would.
- **The algal warning** stays a separate line.

8. Bars

Bar	Selected (agreement-only)	M-iuw ±10 d (not adopted)	Required
Operational	0.693	0.743	9/10: ≥ 0.85; Stage 16: ≥ 7/10
Weakest core class	0.571 (stained)	0.667 (stained)	9/10: ≥ 0.80; Stage 16: ≥ 0.60
Harmful	10.6%	11.6%	9/10: ≤ 5%; Stage 16: ≤ 15%
Severe	1.5%	1.3%	9/10: ≤ 0.5%; Stage 16: ≤ 2%
Strong share / right	11% / 94%	23% / 95%	9/10: ≥ 50% / ≥ 90%
Stage 16 freeze bars	**no** (stained 0.571)	**yes, on development**	
True 9/10	**no**	**no**	

- **The selected variant is Stage 16's answers with better evidence labels.** It meets neither bar.
- **The combining method meets the Stage 16 freeze bars on development, but not the 9/10 bar.** Stained (0.67) is still far from 0.80, harmful far above 5%, and Strong far below 50%.

9. Certification (none here)

What exists. Stage 16's 108 fresh rows are historical context only: 7/10 operational, insufficient. They were not used.

What a Stage 17 design needs. A completely unseen pool. The immediate sources for US stained and limited reservoir Secchi are:

1. **Oklahoma Water Resources Board, Beneficial Use Monitoring Program.** About 130 reservoirs sampled quarterly on a five-year rotation, many of them turbid. Data are served through the University of Oklahoma OKH2O water-quality portal (OWRB BUMP: `oklahoma.gov/owrb/data-and-maps/beneficial-use-monitoring-program.html`, OU portal: `ou.edu/okh2o/monitoring/data-portal/water-quality-data.html`).
2. **Texas TCEQ SWQMIS.** Reservoir Secchi from TCEQ and the river authorities, downloadable directly from TCEQ ahead of WQP's lag (TCEQ SWQM data: `www.tceq.texas.gov/waterquality/data-management/swqm_data.html`).
3. **TVA reservoir monitoring and USACE district water-quality programs:** a direct records request. There is no public bulk download.
4. **A WQP re-pull around April 2027** for the 2026 state uploads.

Before anything is retrieved, each needs the Stage 16 protocol: sites in no earlier file, a value-blind index, and frozen minima.

MULTI-SCENE CONSENSUS USEFUL: NO (by the frozen rule. Every combining method raised the harmful rate and was excluded; the selected agreement-only variant keeps Stage 16's answers, so stained is unchanged. The best combining method, inverse-uncertainty weighting over ±10 days, raised stained by +0.096 (CI +0.054 to +0.143) and operational by +0.050 (CI +0.035 to +0.066), with harmful +0.95 points (CI −0.19 to +2.08): useful on the evidence, but not adopted)

BEST CONSENSUS METHOD: Agreement-only evidence upgrade (±3 days, rainfall rule on, S2C off), the simplest pre-registered variant that does not raise harm. If a ~1-point harmful increase is accepted: inverse-uncertainty weighting of frozen scene predictions over ±10 days without the rainfall rule (M-robust and M-bayes add nothing over it)

STAINED SCORE: 0.571 selected (unchanged from Stage 16); 0.667 with inverse-uncertainty weighting ±10 d (not adopted)

LIMITED SCORE: 0.676 selected; 0.710 with weighting ±10 d

OPERATIONAL SCORE: 0.693 (7/10) selected; 0.743 (7/10) with weighting ±10 d

HARMFUL RATE: 10.6% selected; 11.6% with weighting ±10 d

SEVERE RATE: 1.5% selected; 1.3% with weighting ±10 d

STRONG EVIDENCE SHARE: 11% of usable answers selected; at most 24% for any variant (Stage 16: 11%); the 50% target is not met

STRONG EVIDENCE ACCURACY: 94% selected; 95% with weighting ±10 d

S2C AGREEMENT CAN UPGRADE AUTHORITY: NO (development holds almost no Sentinel-2C: 0 upgraded answers against the 30 needed; testable only on 2025–2026 data)

DEVELOPMENT CAPABILITY: 7/10

TRUE 9/10 DEVELOPMENT BAR PASSED: NO

NEXT UNSEEN CERTIFICATION SOURCE: Oklahoma OWRB Beneficial Use Monitoring Program reservoirs (OU OKH2O portal) and Texas TCEQ SWQMIS reservoir Secchi, retrieved directly for sites in no earlier file under the Stage 16 protocol. Then a TVA/USACE records request and a WQP re-pull around April 2027

READY TO MOVE TO WATER TEMPERATURE: NO (the 9/10 bar is not passed; the selected variant misses the Stage 16 bars (stained 0.571); the combining method meets them on development, but no unseen certification pool exists yet)